Which Reroute Will an Airline Accept? Learning Route Preferences from Revisions

Ramon Dalmau[1], David Perez[2], Franck Ballerini[1], Seddik Belkoura[2],
Vincent Taverniers[2], Gratiel Marius Marin[2], Robin Deransy[1], Benjamin Cramet[3] and Grigori Gustin[3]
EUROCONTROL
[1]Innovation Hub, Brétigny-sur-Orge, France [2]Maastricht UAC, the Netherlands [3]Headquarters, Brussels, Belgium

***Abstract*—Every revision of a flight plan records a choice that an airline actually made, namely the route it dropped and the route it filed instead. Eighteen months of European traffic contain 1.48 million such pairs, a stronger record of preference than earlier work had, whose alternatives were generated rather than filed. This paper fits a ranking model on the 1 134 000 pairs of the training period, describing each route by how much more it costs than the other route of its pair. On three later months the model never saw, it identifies the route the airline chose 68.1 % of the time. Cost rules fail here: preferring the route with less planned fuel scores 45.0 %, so the strongest simple rule is its inverse at 55.0 %, thirteen points below the model. The model is also informative about its own uncertainty, being right 94.9 % of the time on the fifth of cases it is most confident about, so a channel can be aimed at a precision fixed in advance. Aimed at 80 % on the held-out quarter, such a channel reached 79.2 % and surfaced 56 162 proposals carrying a planned-fuel saving of 20.1 kilotonnes that the airlines' own later filings confirm, or 220 to 255 kilotonnes of CO_2 a year. The archive is built from flights that changed route, however, so beyond 6 672 pairs in which a flight kept its route the model is untested on the majority that stay, and it may propose change more readily than airlines would accept it. Even so, what an airline will accept can be read from what it has already filed.**

***Index Terms*—air traffic flow management, airline preferences, learning to rank, revealed preference, fuel efficiency, flight planning**

I. Introduction

On a spring morning in 2026, an Airbus A320 was filed from Amsterdam to Barcelona on a route carrying 102 minutes of attributed air traffic flow management (ATFM) delay, and the operator subsequently re-filed. The new route shed that delay completely, at a cost of 39 minutes of flight time and 1 397 kg of extra planned fuel. Both routes are held in the network's records and so, implicitly, is the operator's judgement that the delay was worth more than the 39 minutes and 1.4 tonnes of fuel that avoiding it consumed.

That judgement is the quantity this paper models. A flight plan is a negotiated object, in that an airline files, the network responds with regulations and delay, and the airline may file again, so that each revision records the route replaced and the route that replaced it.

The operational interest is concrete, because the extra distance that European traffic flies is measured and published every year. The Performance Review Commission reports how much further than the shortest available track the traffic actually flies [1], and this archive shows one of the behaviours behind that extra distance directly, in that airlines frequently revise onto routes that *cost* fuel in order to escape delay. That fuel-cheaper alternatives also go unfiled for want of being noticed in time is a premise of this work rather than a finding of it. Candidate routes are not hard to come by; what is hard is knowing which candidate is acceptable to this particular operator, on this day, with this rotation to protect, and knowing it inside the planning window rather than under tactical time pressure.

The contribution of this paper is threefold. First, it shows that airline route acceptance is learnable at network scale from revision events alone, and that the confidence of the resulting model is calibrated well enough to support a chosen operating point rather than a single accuracy figure. Secondly, it argues that the label construction is itself a design decision worth defending: a revision pair records a preference the airline actually expressed, whereas a generated candidate may never have been evaluated by the operator at all. Thirdly, it reports what the model has actually learned rather than only how well it scores, namely that route structure outweighs any single cost indicator, together with the caveats that such an attribution demands.

The remainder of this manuscript is organised as follows. Section II explains why a pair of routes drawn from a revision constitutes a preference observation and what a model of such observations is for, and Section III positions the work within the literature on rerouting and route choice. Section IV describes the archive, the split protocol and the model, while Section V reports the results. Finally, Section VI concludes and outlines future work.

II. Background

This section explains how a flight-plan revision becomes a labelled preference pair, why that label is preferred to one built from generated candidates, and what the model is for. A reader already familiar with the filing process may move on to Section II-B.

A. Flight-plan revisions as preference data

In the European network a flight plan is filed before departure and may be revised until the flight leaves. Each filing is a separate message and the network keeps the whole sequence, so there is no single record of a route change. What exists is a message that changed the horizontal route, together with the message that preceded it. Those two describe a decision: the operator had committed to the earlier route and then filed the later one for the same flight on the same day, so the later route is the one it chose and the earlier one the route it abandoned.

When the traffic expected in a volume of airspace exceeds what the controllers there can safely handle, a *regulation* is placed on it and the flights crossing it are given calculated take-off times at the rate it can absorb; the difference between a flight's calculated and requested take-off times is its attributed ATFM delay. A lightly binding regulation leaves most of its traffic undelayed, which is why a route can carry a regulation and no delay at all. An operator facing an attributed delay has a choice that is genuinely economic: accept it, or file a route that avoids the constrained volume at the cost of distance, fuel and en-route charges, and neither answer is universally correct. For instance, a flight with a tight aircraft rotation may find forty minutes of delay far more expensive than a tonne of fuel, whereas the same delay on the last leg of the day may well be cheaper than the detour. This is why the decision is worth learning rather than deriving: the trade depends on operator-specific and flight-specific costs that the network does not observe. Published European reference values [2] give the order of magnitude of the cost of delay and are widely used for analysis, but they are not the cost function any particular operator applies.

The same events also settle how the training labels are built, because which of two routes an operator prefers cannot be asked at network scale but it can be observed, and the labels used herein follow the logic of revealed preference (i.e., inferring what an actor values from the choice it made when a concrete alternative was available) [3]. The alternative construction, adopted by the authors' own earlier work, ranks the filed plan above generated candidates that the operator may never have seen or evaluated [4].

Ranking the filed plan above such candidates may therefore require only that the model recognise which route looks like a filed plan, which is a different and much easier skill than knowing which route an operator would prefer, and accuracy alone cannot distinguish the two. For instance, a generated candidate may cross airspace that the operator never uses, or follow a track that its own flight-planning system would discard on company policy; telling such a route from a filed plan calls for no knowledge of preference whatsoever. No dataset offers both label sources for the same flights, so the argument is about how the labels are built, not a measured comparison.

Revisions remove that difficulty (Fig. 1), because both routes were filed by the same operator, for the same flight, on the same day, so the route labelled less preferred is not a plausible alternative constructed after the fact but one the airline had committed to and then abandoned. The question also narrows: the model is not asked what an airline will file, but which of two specific routes it prefers, which is a strictly easier question and precisely the one a proposal channel needs answered.

B. Two operational use cases

A model of route acceptance is worth building only if some decision changes when it exists. The first such decision is a proposal channel: propose to an airline a route that is environmentally cheaper than the one it filed and that the model deems likely to be accepted, on the grounds that the operator may not have seen or evaluated that alternative in time, and it is the use case that Sections IV-B and V simulate.

The second use case is overload release: when a traffic volume is overloaded and flights may be rerouted out of it, a channel would have to decide which flights to approach first, and ranking them by whether an acceptable and cheaper alternative exists would relieve the overload while asking less of the operators concerned. Network-side rerouting formulations acknowledge that they lack exactly this input, namely the likelihood that the operator would actually fly the assigned alternative [5], [6], [7], and the model presented herein estimates it, acceptance remaining voluntary throughout.

Neither use case is tested end to end here. This paper demonstrates the preference model on which both depend, simulates the first against the airlines' later filings, and presents the second as something the model enables.

III. Related work

Work on rerouting in air traffic flow management has largely approached the problem from the network side. Over two decades ago, the authors of [5] formulated rerouting as a dynamic network flow problem, and the authors of [6] subsequently added a stochastic model that reroutes flights as capacity evolves. Both optimise an objective defined by the network and assume, in effect, that the resulting assignment is executed, so that the willingness of the airline to fly it is not modelled.

More recently, route choice in Europe was studied as a trade between fuel and en-route charges [8], which establishes that the choice is economic and therefore learnable, but uses no label in which the operator rejected an alternative it had itself filed. The closest work to the present one is that of the authors of [7], who incorporate airline preferences into collaborative rerouting decisions and show that accounting for them changes which reroutes are selected. Those preferences, however, are inputs to an optimisation, whereas this paper learns the preference function itself, at network scale, from what airlines did rather than from what they stated.

The direct predecessor of this work applies learning to rank to flight routes for a different purpose: predicting the flight plans that have not yet been filed, so that traffic demand can be estimated earlier than the filing horizon allows [4]. That work established the framing adopted herein, namely a gradient-boosted ranker that scores routes by their indicators in context, and it reported the very difficulty that motivates the present paper. Its training pairs take the filed plan as preferred and a set of generated proposals as less preferred, which is the construction that Section II argues against.

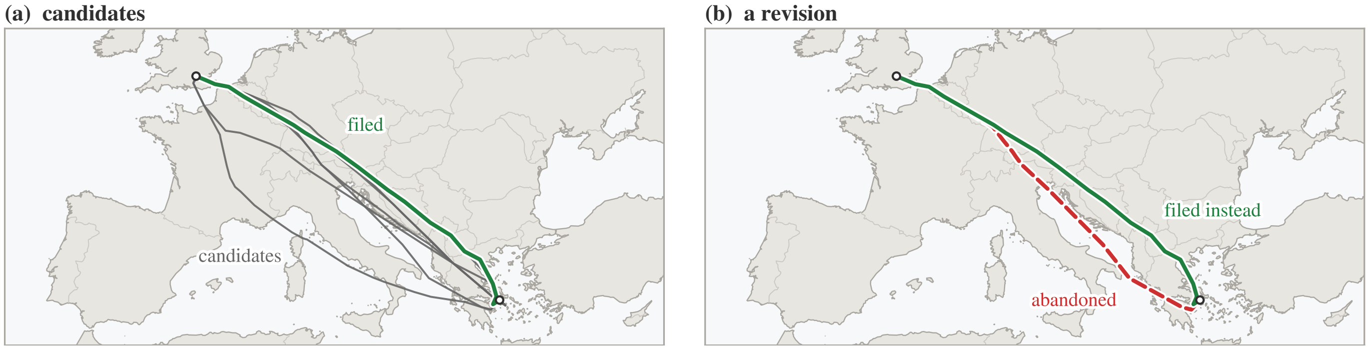

Fig. 1. Two ways to build a training pair, on one city pair. Left: the filed route (solid green) is ranked above candidates that a route generator would supply (grey) and that the airspace user may never have evaluated; the grey tracks drawn here are real routes flown on the same city pair, standing in for generated ones. Right: a revision supplies two routes that the same operator filed for the same flight hours apart, one of which it abandoned. Only the right-hand pair records a preference the airline actually expressed.

IV. Methodology

Figure 2 shows the procedure end to end; the two subsections that follow cover the data and the model. The figure is sufficient on its own, so a reader interested only in the outcome can move on to Section V.

A. Data and pair construction

Eighteen consecutive months of European network data, from January 2025 to June 2026, are used. After retaining only those revisions that change the horizontal route, and only those pairs with complete key performance indicators, the archive contains 1.48 million labelled pairs.

What those decisions look like matters before any model touches them, and Table I breaks the archive down by the regulation context in which each revision was made. Seven revisions in ten are made with no regulation on either route, and the median fuel difference across them is one kilogramme. The pairs that cost fuel lie elsewhere: the revisions that escape a delay outright, or reduce one, together account for 18.4 % of the archive. A revision that escapes a delay moves to a route burning a median 101 kg more than the one it replaced, and that route is the cheaper of the two in only 28.1 % of cases; a revision that merely reduces the delay costs a median 65 kg and is cheaper in 31.6 %. A proposal channel is accordingly not looking for the common case, but for the minority of flights on which a genuinely cheaper alternative exists and would be accepted.

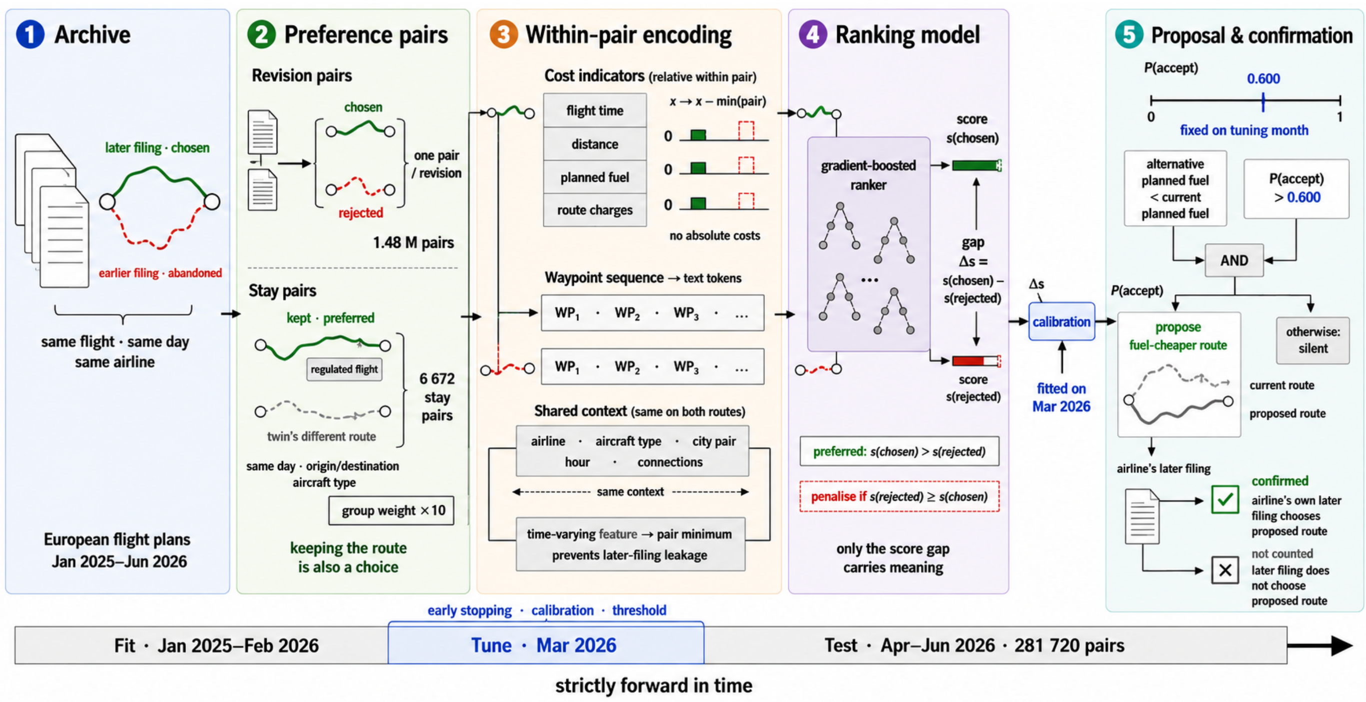

Fig. 2. How a filing archive becomes a proposal, above the fit, tune and test periods. The pair box shows both constructions: a revision, which is one flight and two routes, and a stay pair, which is two flights.

TABLE I

THE REVISION ARCHIVE BY DECISION CONTEXT, OVER ALL EIGHTEEN MONTHS. A ROUTE COUNTS AS REGULATED WHEN A REGULATION WAS ATTRIBUTED TO IT, WHICH LEAVES THE DELAY ITSELF FREE TO BE ZERO. UNREGULATED PAIRS CARRY NO REGULATION ON EITHER ROUTE, ESCAPES CARRY ONE ON THE ABANDONED ROUTE ONLY, THE NEXT TWO CARRY ONE ON BOTH, AND THE LAST ONLY ON THE NEWLY FILED ROUTE. THE FUEL DIFFERENCE IS BETWEEN THE TWO ROUTES OF A PAIR.

context	pairs	share [%]	median Δfuel (new − old) [kg]	new route cheaper [%]
unregulated	1 047 228	70.6	1	46.2
escaped delay	117 594	7.9	101	28.1
reduced delay	155 326	10.5	65	31.6
kept or raised delay	141 648	9.6	0	47.4
took on delay	21 337	1.4	−5	51.8

TABLE II

WHAT THE MODEL SEES FOR EACH ROUTE; SOME ROWS SUMMARISE SEVERAL FEATURES (TWENTY-THREE IN ALL). THE FIRST TWO GROUPS AND THE LAST BELONG TO THE ROUTE AND DIFFER WITHIN A PAIR, THE INDICATORS ENTERING AS DIFFERENCES FROM THE PAIR MINIMUM; THE ROTATION AND CONTEXT ROWS BELONG TO THE FLIGHT AND CARRY THE SAME VALUE ON BOTH ROUTES.

group	content
indicators	flight time, distance, planned fuel, route charges
regulation	attributed delay (own, inbound, outbound), their count, reason, and the type and identifier of the protected volume, meaning the airspace whose capacity the regulation protects
rotation	inbound and outbound connection times and states
context	operator, aircraft type, city pair, hour, weekday, month, time to departure
route	the ordered waypoint sequence, as text

Each route is described by twenty-three features, summarised in Table II. Those two routes are separated in time as well as in geography, however, and that must be addressed before anything is learned. The replacement is by construction the later message, so any feature that drifts with time identifies the answer without saying anything about the quality of the route. The time remaining to departure is the clearest case: it is necessarily smaller for the route that replaced the other, so a model given it in raw form could answer "the one filed later", be right on essentially every pair, and know nothing whatever about routes. This is leakage rather than learning, and it would be invisible in the accuracy figure. To ensure a fair comparison between the two routes, all flight-level attributes were therefore harmonised across the pair so that only route-level quantities differ, and the filing-time feature was collapsed to the pair minimum, so that both routes carry the smaller of the two values and the shortcut disappears.

Two feature-construction decisions shaped the result more than the choice of learner did, and the first is that the four computed indicators enter as within-pair differences rather than magnitudes, obtained by subtracting the pair minimum from each, because absolute quantities are close to meaningless across the network: a 2 000 kg fuel figure means one thing on a short-haul leg and quite another on a long-haul sector, whereas the 300 kg by which one route exceeds the other carries the same meaning whether the sector is 400 or 4 000 km. Secondly, the ordered waypoint sequence is carried as text and encoded as a bag of tokens (i.e., each waypoint identifier is treated as a word and the route as the unordered collection of its words), which lets the model exploit route structure rather than aggregate distance alone. For instance, two routes on the same city pair may differ by under one per cent in distance and in planned fuel, and so be indistinguishable on every indicator, while only one of them crosses a volume whose identifier the model has met in thousands of earlier revisions; the token set carries that fact and the aggregate distance cannot. Section V-B shows that the first of the two matters far more than the authors expected.

Every revision pair is, by construction, a flight that *did* change route, and that creates a trap. A model trained only on such events learns that change is what happens, and recommends it far too eagerly against traffic that mostly stays, a bias that Section V-B measures. The archive can say which route a flight moved to, and nothing at all about the route a flight declined by staying where it was, because staying leaves no record of the alternative, so that record has to come from somewhere else.

A second, much smaller dataset was therefore constructed, in which *keeping* the route is the revealed choice. Let us consider two flights on the same day, with the same origin, destination and aircraft type, that flew different routes and neither of which revised its plan, and let us suppose that one of them was caught by a regulation while the other was not. The regulated flight had both a reason to move and a demonstrated alternative, namely the route that its twin was flying that same day, and yet it stayed. Its route is accordingly labelled as preferred, and that of its twin as the alternative it declined. The construction carries a confound that has to be stated with it: the regulated flight is always the one whose route was kept, so regulation status identifies the labelled answer. The conditions are restrictive and the yield is correspondingly low: 6 672 such *stay pairs*, of which 1 093 fall in the test months, against 1.48 million revision pairs. The two are used together, with stay pairs entering the fitting set at an increased group weight (the pair's weight in the ranking loss).

A rule as crude as "prefer the route of the flight that was regulated" is therefore right on every stay pair by construction, without examining a route at all. Whether the model takes that shortcut can be checked, because the archive also holds the mirror population: in an escape the regulated route is the one *abandoned*, so there the same rule is always wrong, and Section V-B puts the two side by side.

The evaluation is strictly out of time, in the sense that the model is fitted on January 2025 through February 2026 (1 134 064 pairs), tuned on March 2026 (67 349), and evaluated once on April to June 2026 (281 720); the three account for all 1.48 million pairs of the archive. Every threshold, calibrator and stopping decision is taken on the tuning month alone, and flights appearing in the tuning or evaluation months are removed from the fitting set. All the data used is held by the network for operational purposes and was processed under existing data-handling arrangements, so operator identity enters the model as an anonymous code only and no operator or airspace volume is named anywhere in this paper.

B. Model, training and calibration

The operational question is comparative: given two routes for one flight, which of them does the airline prefer? Accordingly, the model gives each route a single number, which is not a probability and has no units, and which is only meaningful next to the number given to another route for the same flight. Training pushes those numbers apart in the right direction, in that for every historical pair the model is penalised according to how far the abandoned route's score sits above the chosen one's, and the penalty shrinks as the gap between them grows without ever reaching zero [9]. Nothing in the objective asks the model to reproduce a route in isolation, which is what makes the pair the natural unit.

Gradient-boosted decision trees are used as the scoring function [10], at a depth of 8, with the learning rate selected automatically and with early stopping on the tuning month. Stay pairs enter the fitting set with a group weight w, swept over $\{10, 30, 100\}$.

Once trained, the model has still to be connected to a decision, since a bare score gap of 3.7 in the model's own arbitrary units conveys nothing to an operations room and has to be turned into a statement of the form "about eight times in ten, the route the model prefers is the one the airline goes on to file". That conversion is what calibration provides, namely a monotone mapping from score gap to the observed rate at which the preferred route was the one filed, fitted on the tuning month alone [11]. Isotonic regression was selected there on log loss, against the raw sigmoid of the score gap and a two-parameter Platt fit, before the evaluation months were touched. Models of this kind usually over- or understate their confidence [12], and this one runs the other way, in that on the held-out quarter its own raw sigmoid was better calibrated than the isotonic fit chosen to correct it, missing the observed rate by 0.9 percentage points on average against 1.8. The selection stands as made, so on this archive the fitted mapping is a safeguard rather than a correction, and it is that 1.8-point agreement between stated and observed rates on which a precision target rests: without calibration one can rank proposals but cannot decide how many of them to make. Under the resulting policy a flight is eligible whenever its two routes differ in planned fuel at all, which holds for 97.9 % of the held-out pairs, and the cheaper of the two is the candidate. It is proposed when its calibrated acceptance probability exceeds a threshold τ selected on the tuning month for a target precision and reported unchanged on the evaluation months. Since the calibrator scores the route the model prefers, the channel compares one minus that probability when the cheaper route scores lower, and then stays silent at any threshold of a half or above. A proposal is realised when the airline's own subsequent filing chose that route. Note that the cheaper route is not usually the one the airline went on to file: in 55.1 % of eligible pairs it is the route being abandoned, so a channel proposing the fuel saver on every eligible flight would be right

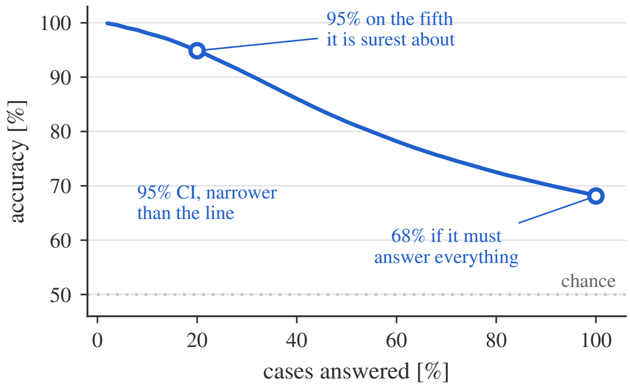

Fig. 3. Accuracy against coverage on the held-out quarter, a case being one pair of routes. Moving left, the model answers fewer cases, lowest confidence dropped first, and is right more often; the dotted line is chance.

44.9 % of the time, which is worse than a coin toss. What the channel is worth is therefore the distance between that 44.9 % and what the model delivers.

V. Results

The model is reported first on how often it is right, then on which choices in its construction earned that, and last on one proposal followed from end to end.

A. Predictive performance

On the held-out quarter the model identifies the airline's chosen route in 68.1 % of 281 720 pairs (95 % interval 67.9–68.3, widening to 67.7–68.6 when flights on the same city pair are counted as one correlated group), against the 50 % of random choice, and reassuringly the tuning month scores 67.5 %, so there is no sign of overfitting to it. Simpler methods on the same quarter are the more informative reference: every rule preferring the cheaper route scores below chance, the fuel rule at 45.0 %, so the strongest single-feature rule is its inverse, "prefer the route that burns more planned fuel", at 55.0 %; the best rule anyone would deploy, preferring the route with less attributed delay, is right 71.6 % of the time but can decide only the 14.6 % of pairs on which both routes carry a delay figure at all, an unregulated route having none rather than a zero, and the two differ, so that scoring the remainder as coin tosses gives it $0.146 \times 71.6 + 0.854 \times 50 = 53.2\,\%$ across the archive.

Accuracy alone understates operational value, because the model is informative about its own uncertainty. Fig. 3 reports accuracy when the model is permitted to abstain on those cases in which the two routes score close together. Let us call the share of pairs that the model is allowed to answer its coverage, the remaining pairs being left alone. On its most confident fifth the model is right 94.9 % of the time, and a channel need not answer every flight.

A margin of thirteen percentage points over the best single-indicator rule establishes that route choice carries structure that no one indicator recovers, which was by no means self-evident, and the delay rule is the instructive case: it is accurate on the pairs it can decide and silent on the remaining 85.4 % of the

TABLE III

ACCURACY BY STRATUM ON THE HELD-OUT QUARTER, WITH WILSON 95 % INTERVALS. THE THREE BLOCKS SPLIT THE SAME 281 720 PAIRS THREE WAYS, THE LAST OF THEM BY HOW MANY OF THE THREE COSTS AN AIRLINE PAYS FOR DIRECTLY, NAMELY FLIGHT TIME, PLANNED FUEL AND ROUTE CHARGES, FAVOUR THE NEWLY FILED ROUTE.

stratum	pairs	accuracy [%]
which route carried a regulation		
neither	178 370	64.0 [63.8, 64.2]
old route only	30 953	88.0 [87.6, 88.3]
both	67 587	70.6 [70.2, 70.9]
new route only	4 810	58.6 [57.2, 60.0]
delay on the abandoned route		
none	205 205	63.9 [63.7, 64.1]
up to 15 min	19 672	70.3 [69.7, 71.0]
15 to 30 min	18 283	79.3 [78.7, 79.9]
30 to 60 min	21 231	84.5 [84.0, 85.0]
above 60 min	17 329	83.5 [82.9, 84.0]
cost indicators favouring the new route		
0 of 3	90 961	79.9 [79.7, 80.2]
1 of 3	87 471	70.0 [69.7, 70.3]
2 of 3	78 552	56.8 [56.5, 57.2]
3 of 3	24 736	53.9 [53.3, 54.5]

archive, which is exactly the gap a learned model closes. That margin does not establish, however, that 68.1 % is anywhere near the ceiling of this task. Part of the residual error falls on pairs in which the two routes are nearly indistinguishable on every recorded quantity, and part on decisions driven by information that no route-level dataset contains (e.g., slot swaps, crew legality, curfews or company routing policy). Neither share is separable from genuine model error without data that the network does not hold.

Where those 68.1 % sit is more informative than the pooled figure, and Table III splits the quarter three ways. Accuracy climbs steeply with the delay carried by the abandoned route, from 63.9 % where there was none to about 84 % beyond half an hour, which is what an economic reading predicts: the larger the delay, the clearer the incentive, and the easier the decision is to predict. The model is accordingly strong precisely where an airline is trading fuel against delay and the stake is large, and weak precisely where the two routes cost almost the same and the choice carries little consequence either way. The third block of the table is the one that repays attention, because it inverts the expectation. When none of the three costs an airline pays for directly, namely flight time, planned fuel and route charges, favours the newly filed route, so that the airline moved onto a route worse on every recorded cost, the model still identifies the move 79.9 % of the time, whereas when all three favour it accuracy falls to 53.9 %. The two results are in fact one result, because a move against every indicator is almost always a move with a strong non-cost reason, typically a regulation, and such reasons are visible to the model, whereas a move that merely tidies up a route the indicators already preferred carries little signal about *when* the airline bothers to make it. The model contributes most where cost logic fails, which is the right division of labour for a system meant to complement cost models rather than replace them.

TABLE IV

SCREENING ABLATIONS. EACH VARIANT IS RETRAINED FROM SCRATCH AT A REDUCED SETTING (DEPTH 6, 800 ROUNDS) THAT REACHES 60.6 % ON REVISION PAIRS; VALUES ARE THE CHANGE IN REVISION ACCURACY AGAINST THAT BASELINE, NOT HEADLINE ACCURACIES. A POSITIVE VALUE MEANS THE VARIANT SCORES *higher* ON REVISIONS, WHICH FOR THE STAY-PAIR CORRECTION IS THE PRICE PAID FOR THE CHANGE-BIAS REDUCTION THE TEXT REPORTS NEXT. EVERY DIFFERENCE IS SIGNIFICANT UNDER MCNEMAR'S TEST AT $p < 0.001$ EXCEPT THE WAYPOINT TEXT, WHICH AT THIS REDUCED SETTING APPEARS WORTHLESS ($p = 0.58$) AND AT FULL SCALE IS WORTH 1.4 POINTS. THE MONOTONE CONSTRAINT FORCES THE SCORE NEVER TO RISE WITH THE NUMBER OF REGULATIONS ON A ROUTE; A VERTICAL-ONLY REVISION LEAVES THE HORIZONTAL ROUTE UNTOUCHED, SO ITS TWO ROUTES ARE IDENTICAL AND THE PAIR UNANSWERABLE.

variant	Δ accuracy [pp]
indicators absolute, not within-pair	-10.0
without stay-pair correction	$+3.7$
with monotone constraint	-0.2
without waypoint text	$+0.03$
including vertical-only revisions	-2.0

A model that scores well on average but decays from month to month is not deployable, and accuracy is therefore reported per month. Across the three test months it moves between 67.5 % and 69.0 %, and a replication of the same recipe on 2025 alone scores 67.3 % and 67.5 % on its own two test months. Five months of evidence rule out rapid decay, but they do not establish the retraining cadence that a deployed system would need.

B. Ablation and change bias

Table IV reports screening ablations, each configuration changed in turn and the model retrained.

Expressing the indicators relative to the other route is decisively the most important choice: removing it costs nearly ten percentage points, far more than any other component, because it is what converts four absolute magnitudes into a comparison.

The waypoint-text result requires care, and illustrates why screening ablations can mislead. Interestingly, at screening scale the text feature appears worthless, and yet retrained at full scale keeping it is worth 1.4 percentage points (68.2 against 66.8, the 68.2 being a separate run and within the half point these figures vary by), the reason being that a bag-of-words representation spread over thousands of rare tokens cannot repay its cost in 800 shallow rounds. It is a real but secondary contributor, worth about a tenth of the model's margin over the best single rule.

The correction whose price the table records is aimed at exactly that trap, and its effect is large. A model trained on revisions alone prefers the alternative route in 65.2 % of the cases where the flight demonstrably kept its own, and so is right on the remaining 34.8 %, because revisions teach it that a regulated route is one to escape and it carries that lesson into a population where the opposite holds. Adding stay pairs at group weight 10 turns that 34.8 % into 90.0 %, at a cost of 0.8 points on the revision metric at full scale, 68.1 % against 68.9 % (the 3.7 points of Table IV are measured at the reduced screening setting).

That 90.0 % is not the model reading the confound back off the construction. A rule keyed on regulation status alone would score 100 % on stay pairs and zero on escapes, whereas the same model scores 90.0 % on the 1 093 stay pairs and 88.0 % on the 30 953 escapes of the held-out quarter, which no such rule can do, since telling the two situations apart takes the route and the flight around it. The weight $w = 10$ is used here because it takes nearly all of the available correction at the smallest cost: raising it to 100 reaches 97.6 % on stay pairs at 66.7 % on revisions, so a further 7.6 points of stay accuracy cost 1.4 points of revision accuracy, and 30 lies between the two.

C. Error analysis and feature attribution

Fortunately, errors concentrate where the model is already unsure. Only 5.6 % of them fall among the quarter of pairs whose two routes score furthest apart, against the 25 % that would fall there if error were independent of confidence, and 27.7 % are near-ties in which the two scores land within one fourth of the archive's median gap. Errors therefore concentrate at low confidence, which is what makes confidence filtering work. The confident errors are nevertheless instructive, as one case from the held-out quarter shows: a flight re-filed off a route carrying 124 minutes of attributed delay onto a route with none, saving fuel and time, and the model confidently scored the abandoned route higher. The flight-level context is identical within a pair by construction, and here every route-level quantity the model reads except the waypoint sequence favoured the move, so the preference came from the waypoint sequence, presumably a learned structural preference that overrode the rest. Confident errors are therefore not all of the hidden-information kind, meaning decisions driven by facts that appear in no route-level record, such as a slot swap or a curfew. Opportunely, confident errors of this kind cost less than their share of the error rate suggests: the only route the channel could have offered is the cheaper one, the flight had just filed it, and a deployed channel, which unlike this simulation knows what is on file, would have said nothing at all. The expensive errors are the opposite ones, in which the model confidently proposes a route that the operator declines, and the change-bias measurement above is the closest available proxy for their rate.

A ranking model an operational user is asked to trust must be open to interrogation, and Shapley attributions were therefore computed over the held-out quarter [13]. Because the model decides on a comparison rather than on one route, what Fig. 4 reports is the *difference* between a feature's attribution on the two routes of a pair, averaged in magnitude over the quarter. That is a ranking not of what correlates with acceptance, but of what moves this particular model's decisions.

The waypoint sequence is the single largest term, ahead of every *individual* cost indicator though the four together sum to more, which means that route structure carries information about the choice that the cost indicators do not. Attributed ATFM delay comes next, moving the score more than planned fuel does. What the figure ranks, in either case, is what moves this model's decisions and not what an operator values. One entry is a caution rather than a finding: time to departure appears fourth even though the encoding makes it identical on both routes, so what is measured there is its interaction with the features that do differ.

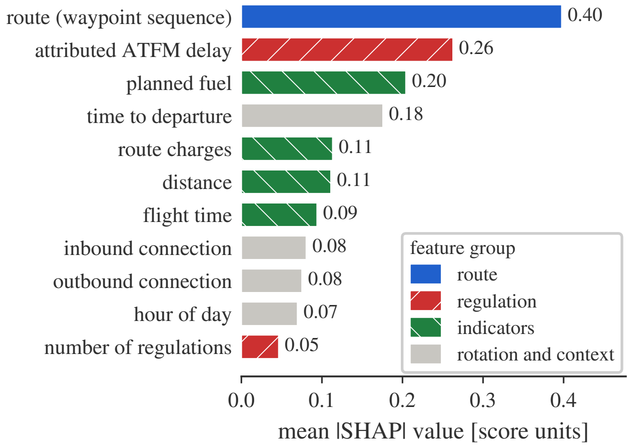

Fig. 4. What moves a decision: the mean absolute *difference* between a feature's Shapley value on the two routes of a pair, over the held-out quarter, with the rotation and context groups of Table II drawn together. The waypoint sequence is the largest single term, ahead of every *individual* cost indicator, though the four taken together sum to more; attributed ATFM delay outweighs planned fuel.

The waypoint text ranks first by attribution, yet removing it and retraining costs only 1.4 points of accuracy. Attribution measures how much a feature moves the score of the model we have; ablation measures how much worse a model without it would be. Where features are correlated, as route geometry and cost indicators plainly are, the two answer different questions and the second is the one that governs what may safely be dropped.

D. A worked proposal and the conditions for deployment

One case from the held-out quarter shows what the channel is for, and is worth following from end to end, because every step is a step that the deployed system would take.

A narrow-body was filed on a medium-haul sector, with no regulation attributed to either route. The system first takes the route then on file and the one alternative offered against it, computes their indicators as differences within the pair, and finds that the alternative burns 2 016 kg less planned fuel, close to a quarter of the trip fuel, at essentially the same flight time. Both routes carry the same flight-level context, so nothing the model sees distinguishes them except where they go and what they cost.

The model then scores each route, and the alternative comes out $+3.7$ ahead on that same unitless scale. That margin is not yet a decision: it passes through the calibrator fitted on the tuning month, which converts it into an acceptance probability, and that probability is compared with the threshold $\tau = 0.600$ fixed before the evaluation period began. This case clears it and the policy therefore proposes; had the margin been smaller the flight would have been left alone, which is what happens on roughly four flights in five.

The operator's own later filing chose exactly that route. Keep in mind, however, that the airline was told nothing, so this is evidence not that a proposal would have been accepted, but that the model identified, before departure, a route the airline went on to prefer. That is the strongest validation available without running the channel, and the form on which every fuel figure here rests.

One proposal says little about a channel making tens of thousands, and three conditions stand between these results and an operational channel.

Data currency binds first, because attributed delay is among the strongest inputs and is a property of a moment rather than of a flight, so what constrains a deployed channel is the freshness of the regulation picture rather than the cost of computation.

The population changes next, in that a deployed channel would see every flight and most never revise their plan at all, so the model would be asked a question on which it was evaluated only through the stay pairs of Section V-B; contentment with the filed plan is only one of the reasons a flight does not revise.

Acceptance itself, finally, is never observed here at all: every figure rests on what the airline filed next, which is the strongest evidence available until proposals are actually made.

At an operating point fixed in advance for 80 % precision ($\tau = 0.600$, delivering 79.2 %), the simulated channel surfaces 56 162 proposals over the held-out quarter carrying 20.1 kt of planned fuel that the airlines' own later filings confirm, or 63.5 kt of CO_2 at 3.16 kg per kilogramme of fuel, and between about 220 and 255 kt a year. These figures are computed on flights that revised their plans, and planned fuel is not burned fuel, so they should be read as what a channel could have surfaced earlier rather than as a traffic-wide counterfactual. The full channel analysis is the subject of a journal extension of this work.

VI. Conclusions

Airline route acceptance is learnable from flight-plan revisions at network scale, and the findings of this paper are threefold. First, a model fitted on 1 134 000 revisions identifies the chosen route in 68.1 % of held-out cases, thirteen percentage points above the best rule built on any one indicator, and supports operation at a selected precision rather than at a single accuracy figure: on the most confident fifth of the decisions it is right 94.9 % of the time. Secondly, much of what the model can learn is decided before training, by the label construction: a revision pair records a preference that the airline actually expressed, whereas ranking a filed plan above generated candidates can be solved by recognising filed plans alone. Thirdly, the model weighs route structure more heavily than any individual cost indicator, though the four together outweigh it, and that structure is information a formulation based on costs alone cannot represent.

The number that characterises this system is not 68.1 % but the curve of Fig. 3, because a channel chooses where on that curve to sit, and because ranking a pair correctly and proposing correctly are different quantities of which the second is the operational one: at the threshold used here, the channel proposes on about a fifth of the flights whose two routes differ in planned fuel at all, and 79.2 % of those proposals match what the airline went on to file. Accordingly, any first deployment would sit well to the left of that curve, and would move right only as acceptance is observed.

One of the deployment conditions set out above also bounds what this work can claim scientifically: the data on flights that stayed is small, 6 672 pairs in all and 1 093 in the test months, and pairs two different flights rather than two routes for one, so a model could learn to recognise the construction rather than the preference it encodes. Future work should therefore be directed, in the first place, at assembling a balanced stay dataset in which regulation status is independent of the label, which requires data that is not currently assembled rather than data that does not exist. Last but not least, a shadow-mode trial logging the real responses of operators would close the remaining gap.

Disclaimer

The views expressed are those of the authors and do not necessarily reflect the official position of EUROCONTROL or of its Member States. This paper reports research and does not commit to any operational service.

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